

## 2. Probability Basics

**Name : Rakesh Ponnoju**

Assignment Link : <https://github.com/RakeshPonnoju0558/DATA-ANALYTICS-PW/tree/master/Assignments/Statistics/2-Probability-Basics>

Mail id : rakeshponnoju6@gmail.com

Question 1: A die is rolled. What is the probability of getting:

**(a) An even number**

**(b) A number greater than 4**

**Solution:**

- Total possible outcomes when a fair die is rolled = 6 (numbers 1,2,3,4,5,6).

**(a)** Even numbers on a die =  $\{2, 4, 6\} \rightarrow \text{count} = 3$ .

Probability  $P(\text{even}) = \frac{\text{favourable outcomes}}{\text{total outcomes}} = \frac{3}{6} = \frac{1}{2} = 0.5$ .

**(b)** Numbers greater than 4 =  $\{5, 6\} \rightarrow \text{count} = 2$ .

Probability  $P(> 4) = \frac{2}{6} = \frac{1}{3} \approx 0.3333$ .

**Question 2: In a class of 50 students:**

**20 like Mathematics (M)**

**15 like Science (S)**

**5 like both subjects**

**What is the probability that a student chosen at random likes Mathematics or Science?**

**Solution:**

We want  $P(M \cup S)$ . Use the union formula:

$$P(M \cup S) = \frac{|M| + |S| - |M \cap S|}{\text{total students}}.$$

Plug numbers:

$$|M| = 20, |S| = 15, |M \cap S| = 5, \text{ total} = 50. P(M \cup S) = \frac{20 + 15 - 5}{50} = \frac{30}{50} = \frac{3}{5} = 0.6.$$

Final answer: **0.6 (or 3/5, 60%)**.

**Question 3: A bag has 3 red and 2 blue balls. If one ball is drawn randomly and is red, what is the probability that the next ball is also red (without replacement)?**

**Solution:**

Initial: 3 red, 2 blue → total 5.

One ball drawn and it is red. Without replacement means that drawn red is not returned.

Remaining: red = 3 − 1 = 2, total remaining = 5 − 1 = 4.

Probability next is red:

$$P(\text{next red} \mid \text{first red}) = \frac{2}{4} = \frac{1}{2} = 0.5.$$

Final answer: **1/2 (0.5)**.

**Question 4: The population of a school is divided into 60% boys and 40% girls. If you want equal representation of both genders in the sample, which method should you use: Simple Random Sampling or Stratified Sampling? Why?**

**Solution:**

- **Answer: Stratified Sampling.**

**Reason (step-by-step logic):**

1. **Goal:** equal representation of both genders in the sample (i.e., sample should include same number or proportionally equal counts from boys and girls as required).
2. **Simple Random Sampling (SRS)** draws randomly from whole population — it may not preserve desired gender proportions (especially if you need exact equal numbers).
3. **Stratified Sampling** divides population into strata (here: boys and girls) and then

samples separately from each stratum. You can then **control** sample sizes from each stratum to achieve equal representation (e.g., sample same number from boys and girls).

4. Therefore, stratified sampling ensures controlled, equal representation by design.

**Final: Use Stratified Sampling because it lets you sample separately from boys and girls and fix equal counts.**

**Question 5: The average height of 1000 students = 160 cm. A sample of 100 students shows an average height = 158 cm.**

**Find the sampling error.**

**Solution:**

Sampling error = (sample statistic) – (population parameter).

Here:

$$\text{Sampling error} = \bar{x}_{\text{sample}} - \mu_{\text{population}} = 158 - 160 = -2 \text{ cm.}$$

Interpretation: sample mean is 2 cm less than population mean. Magnitude = **2 cm**; signed error = **-2 cm**.

**Question 6: The population mean salary is ₹50,000 with  $\sigma = ₹5,000$ . If we take a sample of 100 employees, what is the standard error of the mean (SEM)?**

**Solution:**

Standard error of mean (SEM) formula:

$$SEM = \frac{\sigma}{\sqrt{n}}$$

Given  $\sigma = 5000$ ,  $n = 100$ ,  $\sqrt{100} = 10$ .

$$SEM = \frac{5000}{10} = 500.$$

Final answer: **₹500**.

**Question 7: In a group of 100 students:**

**40 like Cricket (C)**

**30 like Football (F)**

**10 like both Cricket and Football**

**Find the probability that a student likes at least one sport.**

**Solution:**

We want  $P(C \cup F) = \frac{|C|+|F|-|C \cap F|}{total}$ .

Plug in:

$$|C| = 40, |F| = 30, |C \cap F| = 10, total = 100. P(C \cup F) = \frac{40 + 30 - 10}{100} = \frac{60}{100} = 0.6.$$

Final answer: **0.6 (60%)**.

**Question 8: From a deck of 52 cards, two cards are drawn without replacement. What is the probability that both are Aces?**

**Solution:**

- Number of Aces initially = 4.
- Total cards initially = 52.

Probability first card is an Ace:  $4/52$ .

Given first is an Ace, remaining Aces = 3, remaining cards = 51  $\rightarrow$  probability second Ace =  $3/51$ .

By multiplication rule (without replacement):

$$P(\text{both Aces}) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652}.$$

Simplify:

$$12/2652 = 1/221 \approx 0.0045248869.$$

Final answer:  $\frac{1}{221} \approx 0.004525$  (about 0.4525%).

**Question 9: A factory produces bulbs with 2% defective rate. If 5 bulbs are chosen at random, what is the probability that all are non-defective?**

**Solution:**

- Defective rate = 2%  $\rightarrow p(\text{defective}) = 0.02$ .
- So  $p(\text{non-defective}) = 1 - 0.02 = 0.98$ .

Assuming independence, probability all 5 non-defective:

$$P(\text{all non-defective}) = 0.98^5.$$

Compute stepwise:

$$\begin{aligned} 0.98^2 &= 0.9604, 0.98^3 = 0.9604 \times 0.98 = 0.941192, 0.98^4 = 0.941192 \times 0.98 \\ &= 0.92236816, 0.98^5 = 0.92236816 \times 0.98 = 0.9039207968 \approx 0.90392. \end{aligned}$$

Final answer:  $0.98^5 \approx 0.90392$  ( $\approx 90.392\%$ ).

**Question 10: Differentiate between discrete and continuous random variables with examples.**

**Solution:**

- **Discrete random variable:**
  - Takes **countable** values (finite or countably infinite).
  - Values separated by gaps; can list possible outcomes.
  - **Example:** Number of students present in a class (0,1,2,...), number of heads in 10 coin tosses.
  - Probability mass function (PMF) applies.
- **Continuous random variable:**
  - Takes values from an **uncountable** range (any value in an interval).

- Outcomes form a continuum — infinite values between any two points.
- **Example:** Height of a student (e.g., 158.34 cm), weight, time to complete a task.
- Probability density function (PDF) applies; exact probability at a single point is 0, but probability over an interval is positive.

**Concise contrast table:**

| Aspect             | Discrete RV                      | Continuous RV                                     |
|--------------------|----------------------------------|---------------------------------------------------|
| Values             | Countable (integers, categories) | Uncountable (real intervals)                      |
| Probability        | PMF ( $P(X=x)$ )                 | PDF; $P(X \text{ in interval}) = \text{integral}$ |
| Example            | Number of defective bulbs in 5   | Height, weight, time                              |
| Single-point prob. | Can be $>0$                      | Always 0 (needs interval)                         |